Hari Om TubeOS: all worked samples and examples (collected from every chapter)

Accurate as of 31 October 2026. Values are examples, not real customers.

From chapter 00: Welcome to Hari Om TubeOS: the big picture

Sharma Tubes Pvt Ltd sends PO GOLIVE-PO-118 for 5,000 pcs of 76.4 x 81 x 150 at ₹2.75 per pc: 3,000 BLUE and 2,000 RED.

1. Sales enters **SO/26/10/412**; a planner approves it.
2. Sales releases it: two job cards, **26/10/315** (BLUE, 3,000 pcs) and **26/10/316** (RED, 2,000 pcs), go to the winder queue.
3. The planner places 26/10/315 on a winder, Shift A, 31 Oct. Operators log Winding, Oven and Process runs with two QC readings each.
4. Meanwhile the RM inward calendar shows Kraft 300 GSM running short on 4 Nov. The Owner approves an arrival; PO **RP-PM/68** goes to Bharat Paper Mills, arrives as **P/26/014** with 12 reels, passes incoming QC and is issued to the floor.
5. When 26/10/315 closes Process, Dispatch makes the challan for 3,000 pcs; the order shows **Partly dispatched** until 26/10/316 ships too.

From chapter 01: Getting started

Ramesh is a Planner who also covers Sales on Saturdays, so his login has both roles.

- He signs in and lands on the **Planning board** (Planner comes first).
- He opens the role box and picks **Sales**; TubeOS opens **Orders**.
- He presses Ctrl+K, types "calendar" and opens **RM inward calendar**, which he sees because his Planner role has it.
- At 1:30 pm he leaves for lunch. At 1:50 pm the page asks him to sign in again; after signing in he is back on the RM inward calendar.

From chapter 02: Dashboard and inbox

Monday 9:00, the Owner opens TubeOS:

- Headline: "2 overdue job cards, 1 card on QC hold and 4 low-stock materials need attention in Plant A."
- Action center: **Sales orders to approve 3** (SO/26/10/418 Sharma Tubes Pvt Ltd · draft · 5,000 pcs), **Purchase orders to approve 1** (RP-PM/70 Bharat Paper Mills · revision 1 · ₹4,12,500), **RM arrivals to approve 2** (Kraft 300 GSM · 4 Nov · 2,500 kg).
- She approves the SO from its order page, the PO in PO approvals and the arrivals on the RM inward calendar. Back on the dashboard the three lanes show "Nothing waiting".

- Inbox: **Needs action 12**. She opens "Purchase order awaiting approval: RP-PM/70" (already done, so it is just marked read) and presses **Mark all read**.

From chapter 03: Using TubeOS on a phone

Suresh, winder operator, Shift A, 31 Oct:

1. 08:15: he scans the QR on job card **26/10/315** (Sharma Tubes Pvt Ltd, BLUE, 3,000 pcs = 300 bamboo × 10). The card opens on **Winding**.
2. He taps **Log output**: machine SHARMA_WINDER-2, 120 bamboo produced, 118 accepted. Rejected 2 is worked out by the system. He saves; the run goes to the supervisor.
3. He taps **Log QC** for Run 1 and enters two readings for each winding check. One C.S. reading is outside the band: it is saved with a flag.
4. 08:40: store keeper Anil scans reel **AT-26-0412** (Kraft 300 GSM, 412 kg). Reel & coil issue opens with the reel picked; he presses **Issue**. It is recorded as Shift A, 31 Oct, and listed under Slitting cards to write.

From chapter 04: Sales orders

Sharma Tubes Pvt Ltd sends PO 4500012345 dated 28 Oct for 76.4 x 81 x 150 tubes.

Field	Value
Line 1 qty	5,000 pcs at ₹2.75 (last rate ₹2.75 on SO/26/10/388, pressed **Use**)
Colours	BLUE 3,000 pcs; RED 2,000 pcs at ₹2.80
Line value	3,000 × 2.75 + 2,000 × 2.80 = ₹13,850 ex-GST
Delivery date	blank → "No date"
Expiry	blank → 45 days from 28 Oct = 12 Dec

1. Priya (Sales) creates it: **SO/26/10/412**, Draft. She presses **Send for approval**.
2. Ramesh (Planner) presses **Approve**: Ready to release.
3. Priya releases: BLUE 3,000 on SHARMA_WINDER-1, RED 2,000 on SHARMA_WINDER-2. Two job cards, 26/10/315 and 26/10/316. Status: In production.
4. 26/10/315 ships 3,000 pcs: **Partly dispatched**, "3,000 of 5,000 dispatched".
5. The customer now says it does not want the RED pieces. They are already on 26/10/316, so Short close has nothing to drop and explains: "Everything left is already on job cards; the order closes by itself once they are dispatched." To stop the card, the planner force-closes 26/10/316 in the Job card manager. If the order was short-closed first, the balance that card hands back is dropped too instead of becoming releasable again.

From chapter 05: Specifications

Sharma Tubes Pvt Ltd orders 76.4 x 81 x 150 tubes, dry weight 33.78 g, C.S. 400 kgf, 10 pcs per bamboo.

Item	Rest of year	Monsoon
Paper	SHARMA_300 Kraft 300 GSM × 3 plies	SHARMA_300 × 2 plies + Kraft 280 GSM × 1 ply
Paper weight	32.74 g	32.10 g
Adhesive	Glue 100%, 4.22 g	Glue 100%
Revision	r1, approved	r2 (revised 15 Jun, "Higher humidity, lighter outer ply"), approved

- In June the Owner switches the season to **Monsoon**. SO/26/06/120 released on 16 Jun gets job card 26/06/044 with the Monsoon r2 recipe frozen on it.
- On 1 Oct the Owner switches back to **Rest of year**. Card 26/06/044, still open, keeps Monsoon r2. New releases use Rest of year r1.

From chapter 06: A day in the life

Thursday 31 Oct, Plant A:

- 08:10 Priya (Sales) enters SO/26/10/418 for Sharma Tubes Pvt Ltd, 2,000 pcs RED, and sends it for approval.
- 08:30 Ramesh (Planner) approves and releases it: job card 26/10/320 in the SHARMA_WINDER-1 queue; he places it on Shift A.
- 08:45 Anil (RM issue) scans reel AT-26-0415 (cleared) and issues it; it shows under Slitting cards to write.
- 09:00 Suresh (Operator) scans 26/10/320, logs 100 bamboo on Winding, and two readings per check.
- 13:30 Meena (Plant Manager) submits the runs and closes Winding.
- Next day the card closes Process; Ravi (Dispatch) makes the challan for 2,000 pcs.

From chapter 10: Job cards

Sharma Tubes Pvt Ltd orders 900 pcs of 76.4 x 81 x 150, BLUE, on SO/26/10/408. Sales releases it; card **26/10/315** is made with 900 pcs = 90 bamboo × 10 pcs, bamboo 1,540 mm.

Stage	Target	Made	Good	Rejected	Status
Winding	90 bamboo	90	90	0	Done, 1 flagged
Oven	90 bamboo	90	90	0	Done, 1 flagged
Process	900 pcs	893	893	0	Done, 1 flagged
Dispatch	893 pcs	893	893	0	Done

A Process reading of I.D. 76.3 mm against a band of 76.4–76.6 was saved as "Flagged · I.D. 76.3 mm · not held". The card still moved on. Closing Process posted 893 pcs as finished goods; the card showed **Ready to dispatch**, and after the challan was sealed, **Done**.

From chapter 11: Job card manager

Card 26/10/315 for Sharma Tubes Pvt Ltd: 900 pcs, BLUE, scheduled on Winder 1 for 31 Oct Shift A (600 pcs) and Shift B (300 pcs). The customer wants 300 pcs early.

Field	Value
Action	Split into a new card
New card gets	300 pcs
Reason	Customer call-off earlier
Result: 26/10/315	900 → 600 pcs, keeps its plan
Result: 26/10/315-A	300 pcs, planner queue, unplaced
Slot change	31 Oct · Shift B · Winder 1: 300 → off the board
Order line	stays 900 pcs, now on 2 cards

The planner then places 26/10/315-A on Winder 2 for 1 Nov Shift A and prints it.

From chapter 12: Planner

Card 26/10/315 (Sharma Tubes Pvt Ltd, 900 pcs = 90 bamboo, bamboo 1,540 mm) needs about 139 m of winding. The planner drags it to Winder 1, 31 Oct Shift A. Oven opens for planning; she places it on Oven 1, 31 Oct Shift B (90 bamboo). Process opens; she places 900 tubes on Line 1, 1 Nov Shift A.

Stage	Slot	Logged	Result
Winding	31 Oct · Shift A · Winder 1	90 bamboo, 31 Oct Shift A	Done in slot

Stage	Slot	Logged	Result
Oven	31 Oct · Shift B · Oven 1	90 bamboo, 1 Nov Shift A	Ran on another day or shift ("1 day later than plan")
Process	1 Nov · Shift A · Line 1	nothing by 3 Nov 08:00	Not run → back to the queue (Missed slot)

On 3 Nov the operator logs the Process run dated 1 Nov Shift A; the slot is restored and shows **Done in slot**.

From chapter 13: Log output and QC entry

Night shift, card 26/10/315 (Sharma Tubes Pvt Ltd, 76.4 x 81 x 150, BLUE, 10 pcs per bamboo).

Field	Value typed	Result
Time on the clock	01:30, 31 Oct	Date 30 Oct, Shift B (automatic)
Stage / machine	Process / Line 1	Plan match: As planned · 30 Oct · Shift B
Bamboo taken from Oven	9	9 × 10 = 90 pcs possible
Made / Rejected	86 / 2 (code OD)	Good = 84 pcs
Cutting loss	4, "end trimming"	all pieces accounted for (86 + 4)
Saved by	Operator	"Run P#2 sent to the supervisor"

QC on run P#2: I.D. 76.5 and 76.9 mm against band 76.2–76.5. The second reading is red; the operator saves. Result: "QC saved · run P#2 · flagged", "I.D. 76.9 mm (band 76.2–76.5) · S2". The run is accepted; the flag shows on the card and in In-process QC.

From chapter 14: Quality

Bharat Paper Mills delivers 6 reels of Kraft 300 GSM (P/26/014). The paper band is GSM at least 291 (from the master line) and moisture 6–9 %.

Reel	GSM	Moisture	Decision	Result
R1	302	7.1	By readings	Released
R2	298	6.8	By readings	Released
R3	285	7.4	By readings, reason "GSM low on two sheets"	Held

R3 is on hold. The plant head grants a concession for outer plies ("still fit for use"); the FAIL stays on record. Later, card 26/10/315 has a Winding I.D. reading of 76.9 mm against 76.2–76.5: flagged, not held; it appears in In-process QC under I.D., High.

From chapter 15: Month-end close

September 2026, Plant A. Paper limits are standard (7 % wastage, 7 % gap, minimum 5 kg).

Material	Spec	Expected (spec + 7 %)	Used per count	Gap	Allowed	Status
Kraft 300 GSM	2,000 kg	2,140 kg	2,195 kg	+55 kg	± 149.8 kg	Within limit
Kraft 180 GSM	400 kg	428 kg	470 kg	+42 kg	± 30.0 kg	Needs a note
PVA adhesive	120 kg	120 kg	118 kg	−2 kg	± 8.4 kg	Within limit

The plant head writes on Kraft 180 GSM: "Two reels wet at edges, outer plies scrapped on 18 Sep". It becomes **Explained**. With the count approved and the book adding up, the Owner clicks **Close September 2026**. A Process run dated 29 Sep typed on 2 Oct is now refused until the Owner re-opens September.

From chapter 16: Dispatch

Card 26/10/315 for Sharma Tubes Pvt Ltd closed Process with 893 good pcs (released 900). Tube weight on the spec: 33.78 g.

Challan	Pcs	Bundles (100 per bundle)	Net weight	Vehicle	Left to ship
Challan 1	500	5	16.890 kg	GJ15AB1234	393
Challan 2	393	4 (last holds 93)	13.276 kg	GJ15CD5678	0

After Challan 2 is sealed, the card shows **Done** and the order line SO/26/10/408 counts 893 pcs delivered.

From chapter 20: Purchase orders, PO approvals and requisitions

Store makes a PO for Bharat Paper Mills on 31 Oct 2026.

Line	Material	Ordered	Rate	Amount
1	Kraft 230 GSM	2,000 kg	₹44.50 / kg	₹89,000
2	Kraft 300 GSM	1,500 kg	₹46 / kg	₹69,000

Save PO draft gives **RP-PM/54**, Draft, total "2 lines · 3,500 kg · ₹1,58,000". Store clicks **Submit for approval**. The Owner approves it with the reason "Rates as agreed on call". Store clicks **Send to vendor** › **WhatsApp**; the chip reads "Sent on WhatsApp · 31 Oct". Bharat Paper Mills confirms 6 Nov for line 1 and 10 Nov for line 2; Store records both. On 6 Nov the truck brings 2,012 kg of Kraft 230 GSM, which Store receives with **Make GRN** (see the Goods inward chapter).

From chapter 21: RM inward calendar and expected arrivals

On 31 Oct the Planner opens **Paper** on the calendar. **Kraft 350 GSM** shows 0 kg usable, need 317 kg on today (all of it late) and **Short from 31 Oct**. She clicks **Suggest arrivals**; the suggestion is 450 kg on 3 Nov from Bharat Paper Mills (rounded up to the 50 kg order multiple). She adds it as a proposal. The **Waiting for you** tile on the Owner's screen goes to 1.

The Owner clicks **Review** and **Approve all**: "1 arrival approved. 1 PO draft ready to send: RP-PM/55 (Bharat Paper Mills)." The rate comes from the last approved PO, ₹47.25 / kg. Store opens **Scheduled drafts to send**, clicks **Check & send** › **WhatsApp**. Bharat Paper Mills says 4 Nov; Store records it with **Confirm**. On 4 Nov the row is under **Today** in Expected arrivals; GRN clicks **Make GRN**.

Day	Need	Counted arrivals	Balance at day end
Sat 31 Oct	317 kg	0	–317 kg (Short)
Wed 4 Nov	120 kg	450 kg	13 kg

From chapter 22: Invoice differences, debit notes and purchase reports

Bharat Paper Mills delivers Kraft 230 GSM on GRN P/26/021 against RP-PM/54. PO rate ₹44.50 / kg, invoice rate ₹45.25 / kg, weighed 2,012 kg.

Field	Value
Type	Rate
Difference	+₹0.75/kg
Exposure	+₹1,509
Status	Open

The Plant Manager (not the Store user who made the GRN) ticks the row and clicks **Open debit note**: **DN-261106-001**, Draft, ₹1,509. He clicks **Submit**; Accounts clicks **Approve** and **Issue immutable note**. On 20 Nov the vendor gives a credit of ₹1,509; Accounts types 1509 and clicks **Record settlement**. The note shows "₹1,509 / ₹1,509 / ₹0" and status **Settled**.

From chapter 23: Goods inward, GRN register and labels

Bharat Paper Mills delivers on 6 Nov 2026 against **RP-PM/54**, challan BPM/2231, invoice BPM/INV/884 dated 6 Nov.

Vendor reel no.	Net kg	Width	Form
BPM-7781	702	1,000 mm	Reel
BPM-7782	655.5	1,000 mm	Reel
BPM-7783	654.5	1,000 mm	Reel

Total 2,012 kg of Kraft 230 GSM; PO open 2,000 kg, so the strip shows "more than the PO by 12 kg — reason needed below". Store types "Last reels of the order, supplier confirmed extra kg". Invoice rate ₹45.25 / kg against PO ₹44.50 / kg: an amber note shows ₹1,509.00 more. Store validates and posts: **P/26/021**, 3 lots, AT-261106-001 to AT-261106-003, all **Waiting for QC**. A rate case opens for the plant head. Store prints three A4 labels. QC passes the reels the same afternoon and they appear under **Ready for the floor** on the reel issue desk.

From chapter 24: Issue to the floor: reels, coils and materials

Friday 7 Nov 2026, 10:15 (Shift A). Ramesh on the RM issue desk scans three reels of Kraft 230 GSM from GRN P/26/021 that QC passed yesterday, and one coil.

AT number	Form	Weight	Went to
AT-261106-001	Reel	702 kg	Slitter
AT-261106-002	Reel	655.5 kg	Slitter
AT-261106-003	Reel	654.5 kg	Slitter
AT-261104-011	Coil	98 kg	Winders

He clicks **Issue 4 to floor · 2,110 kg**. At 13:00 Ramesh writes the slitting card for AT-261106-001 from the slitter's figures: 7 coils, 12 kg wastage. Ramesh then issues 25 kg of PVA adhesive (oldest lot A/26/005) and 400 pcs of Box R-150 to the floor. In **Issue history** for 7 Nov, Shift A, the six issues appear with Ramesh's name and the racks they came from; the CSV matches the stock book's **Issued** column.

From chapter 25: Stock counts, stock book, locations and opening stock

Month-end count of October 2026, Plant A. On 29 Oct the store opens the October card (it opened on 29 Oct, the last 3 days) and clicks **Start the October count**. It becomes CNT-20261029-001 with 12 racks plus "Not in any location".

Rack	Result after Rack done
RM-STORE	36 reels found; reel AT-261020-014 (412 kg) not found
CHK-A-02	lot IN/26/031-2 of PVA adhesive: book 60 kg, counted 55 kg
CK3-R1	reel AT-261011-006 found, but the system says it was issued on 14 Oct
CK2-R1	one bag of BLUE parchment with no label, 25 kg

The count is sent to the Owner on 31 Oct. The Owner chooses **Leave as used** for AT-261011-006 and clicks **Approve and post**. Voucher VAR-20261031-001 posts three lines dated 31 Oct 2026: −412 kg on AT-261020-014, −5 kg on IN/26/031-2, and +25 kg as a new lot on QC hold for the parchment, whose label is then printed. The count becomes **Approved · month certificate** for October 2026. The stock book's **Corrections** column for October now includes −392 kg. The Owner then closes October on **Month-end close**.

From chapter 26: Inventory, material items, manual FG, stock alerts and RM costing

Alert policy for Kraft 180 GSM. Store keeper Ramesh drafts: Safety 1,600 kg, Reorder 1,700 kg, Target 2,000 kg, Recovery margin 100 kg, Lead days 7, Minimum order 1,000 kg, Cooldown 24 hours, Recipients Owner,Planner, reason "Winder 2 runs 180 GSM daily from November". The preview reads "Critical at ≤ 1,600 kg · warning at ≤ 1,700 kg · recover above 1,800 kg". Plant Manager Suresh activates it.

Day	Free stock	Evaluate result
3 Nov	1,650 kg	New episode, WARNING (threshold 1,700 kg)
4 Nov	1,420 kg	Same episode, now CRITICAL (threshold 1,600 kg); Planner acknowledges
6 Nov	1,760 kg (GRN from Bharat Paper Mills, part cleared by QC)	Still open: 1,760 is not above 1,800

Day	Free stock	Evaluate result
7 Nov	3,240 kg	RECOVERED

Standard cost for Kraft 220 GSM. Base ₹42.50/kg, Discount 2%, Freight ₹1.20/kg, Unloading ₹0.15/kg. Draft landed = 42.50 × 0.98 + 1.20 + 0.15 = ₹43.0000/kg. If the current active cost is ₹41.80/kg, the 10,000 kg impact shows +₹12,000.

From chapter 27: Reports

Monday review at Plant A, period 1–31 Oct 2026.

Report	Figure	What it means
Planning performance	Done in slot 68.8% (44 of 64 finished slots)	About two in three planned slots ran fully in their own shift
Planning performance	Missed slots 25	25 slots had no entry for 36 h; those cards are back in the queue
Plan vs actual	Run before planning 423 of 520	Most runs were logged before the planner placed the stage
Slitting report	6 carded reels, 3,575 kg issued, 67.5 kg wastage	Wastage % = 67.5 ÷ 3,575 = 1.9%
Customer 360	Sharma Tubes Pvt Ltd: 1 delayed order, OTIF 80%	Watch (not Critical, because OTIF is not below 75%)
Inventory intelligence	Kraft 180 GSM: 2,400 kg available, 9,000 kg issued in 30 days	9,000 ÷ 30 = 300 kg/day; 2,400 ÷ 300 = 8 days → watch

From chapter 28: Masters

New customer for the challan. The Admin adds Sharma Tubes Pvt Ltd: Code WB-SHARMA, GSTIN 24AAKCS4821M1Z3. The system fills PAN AAKCS4821M and the billing state Gujarat (24). Billing address: Plot 14, GIDC Phase II, Vapi 396195. The Dispatch (ship-to) address is their Daman warehouse with stores contact Mr Patel, 98250 12345. The row now shows **Ready**. When Dispatch makes a challan for SO/26/10/412, it prints the GSTIN and the Daman ship-to address.

Paper edit. The Admin changes paper 350 (Kraft) bulk factor from 1.55 to 1.50, so the derived thickness goes from 0.5425 to 0.525 mm. Recipe version 3 of spec 76.4 x 81 x 150 for Sharma Tubes, approved last week, keeps 1.55 in its layers. The next recipe version saved will use 1.50.

Order close reason. The Owner adds "Customer moved the plant" and the code becomes SO-CUSTOMER-MOVED-THE-PLANT. Later the Owner renames it "Customer shifted the plant"; the code stays the same, so orders already closed with it still show it.

From chapter 29: System, audit, production season and the guide

New store keeper. The Admin adds Ramesh Patel: login ramesh.store@hariom.com, a 14-character password with upper and lower case letters, a number and a symbol, roles **Store** and **Inward QC**, primary plant Plant A, allowed plants Plant A. Ramesh signs in and lands on his Store work page. He can raise stock corrections and inspect incoming lots, but he cannot see System or approve counts.

Who changed the access? Next week Ramesh suddenly cannot open Incoming QC. On **Audit** the Owner sets **7 days**, picks Module **Users & access** and searches "ramesh". The user-updated event shows under **What changed** that the Inward QC role was removed, by whom and when. The Owner opens the user, ticks Inward QC again and saves; Ramesh signs in again with the role back.

Monsoon switch. On 1 Jun the Owner clicks Monsoon. The preview shows 41 specifications ready, 3 requiring attention ("Approved season recipe missing") and 18 released cards keeping their contracts. The Owner ticks the acknowledgement, types "Monsoon from 1 June as per QC" and confirms. New releases now use the Monsoon recipes and QC bands; the 3 blocked specs cannot be released until QC and Design finish their monsoon setup.